

There is no pivot position in the third row
 $\therefore$ The columns of B do not span $\mathbb{R}^3$. $\square$

31) Let A be a 3×2 matrix. Explain why the equation $Ax=b$ cannot be consistent for all b in $\mathbb{R}^3$. Generalize your argument to the case of an arbitrary A with more rows than columns.

Solution: Let, $A = \begin{bmatrix} a & b \\ c & d \\ atc & btd \end{bmatrix}$ be a 3×2 matrix, where a, b, c and d are arbitrary real numbers. $a \neq 0$. Assume, $ad \neq bc$.

Claim: The matrix A does not have a pivot position in every row.

Proof: By defn, the pivot position of a matrix corresponds to the leading 1 of a row of the matrix in reduced row echelon form.

$$\begin{aligned} & \begin{bmatrix} a & b \\ c & d \\ atc & btd \end{bmatrix} \xrightarrow[\text{Row 1} \times \frac{c}{a}]{\text{Row 2} = \text{Row 2} - \frac{c}{a} \text{Row 1}} \begin{bmatrix} a & b \\ 0 & d - \frac{bc}{a} \\ atc & btd \end{bmatrix} \xrightarrow[\text{Row 3} = \text{Row 3} - \frac{c}{a} \text{Row 1}]{\text{Row 3} = \text{Row 3} - \text{Row 1}} \begin{bmatrix} a & b \\ 0 & d - \frac{bc}{a} \\ 0 & d - \frac{bc}{a} \end{bmatrix} \\ & \xrightarrow[\text{Row 1} \times \frac{1}{a}]{\text{Row 2} = \text{Row 2} \times \frac{1}{(d - \frac{bc}{a})}} \begin{bmatrix} 1 & \frac{b}{a} \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \xrightarrow[\text{Row 1} = \text{Row 1} - b \times \text{Row 2}]{\text{Row 1} = \text{Row 1} \times \frac{1}{a}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

Verification of whether the final matrix is in reduced row echelon form

- Row 1 and Row 2 are non-zero rows. Row 3 is a zero row. $\therefore$ The condition that all the non-zero rows must be above the zero rows is True.
- Row 1 Leading Entry = 1 in Column 1
 Row 2 Leading Entry = 1 in Column 2
 $\therefore$ The condition that for each row, its leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.
- Row 2 Column 1 = Row 3 Column 1 = 0
 Row 3 Column 2 = 0
 $\therefore$ The condition that for each column having a ~~pivot position~~ leading entry, all positions below the leading entry must be 0 is True.